

Variables & Selection

Variables

A variable in programming is like a variable in math class. It is a name that holds a value. You can think of it like a box that you store something in. If you put candy in your box, then when you open it, there will be candy inside. You can change the value in your variable. e.g.

```
from mckPy import *

x = 0
while running():
    fill("skyblue")
    x = x + 5
    drawRect("red", x, 200, 50, 100, outline=2)
```

In this example, **x** is the variable. It starts with the value 0 (`x = 0` sets the value to 0.) Each time around the while loop the value increases by 5 (`x = x + 5`). You can read this as "x becomes what it was plus 5"

Selection (if)

An if structure allows you to perform commands only when a certain is true. e.g.

```
from mckPy import *

x = 0
while running():
    fill("skyblue")
    x = x + 5
    if x > 800:
        x = 0
    drawRect("red", x, 200, 50, 100, outline=2)
```

This program moves the box to the right, then when it goes off the screen it resets the position to the left.

The key to this example is:

```
if x > 800:
    x = 0
```

This works the same way an if works in English. "If it is raining, bring an umbrella."

Else

You can also specify what to do if your condition is false.

```
from mckPy import *

while running():
    fill("skyblue")
    x,y = mouseX()

    if x < 400:
        drawRect("green", x, y, 50, 50)
    else:
        drawRect("red", x, y, 50, 50)
```

This example will draw a rectangle at the position of the mouse. It will be green on the left side of the screen, red on the right.